

Multiple Choice

1. **Answer:** A

Options: A) 0.3 liters; B) 2 liters; C) 0.2 liters; D) 3 liters; E) 0.03 liters

Note: Correct option: A - 0.3 liters

2. **Answer:** B

Options: A) 75.0 mL; B) 90.0 mL; C) 60.0 mL; D) 120 mL; E) 30.0 mL

Note: Correct option: B - 90.0 mL

3. **Answer:** D

Options: A) 0.1645 amu; B) 0.1987 amu; C) 0.1312 amu; D) 0.1588 amu; E) 0.1756 amu

Note: Correct option: D - 0.1588 amu

4. **Answer:** E

Options: A) 22,102.28 liters air; B) 58,650.00 liters air; C) 49,350.25 liters air; D) 65,292.15 liters air; E) 39,175.40 liters air

Note: Correct option: E - 39,175.40 liters air

5. **Answer:** B

Options: A) 2,3-dimethylbutane < neopentane < n-hexane < n-undecane < methane; B) methane < 2,3-dimethylbutane < neopentane < n-hexane < n-undecane; C) n-undecane < n-hexane < neopentane < 2,3-dimethylbutane < methane; D) neopentane < 2,3-dimethylbutane < methane < n-undecane < n-hexane; E) 2,3-dimethylbutane < methane < n-hexane < neopentane < n-undecane

Note: Correct option: B - methane < 2,3-dimethylbutane < neopentane < n-hexane < n-undecane

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'NH₄⁺'.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Hydrogen diffuses 4 times faster than oxygen'.

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** 305 g/mole. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** 7.3×10^{31} molecules. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key